



E. Master Evenius and Oddius

CPU: 1.00s

Memory: 2048MB

In the mystical land of Numeria, two legendary masters, **Evenius** and **Oddius**, are locked in an eternal duel to determine who will claim the throne of the Stone Kingdom. Their battleground is a pile of stones. This pile has N **magical stones**. Each stone possesses an ancient power, and only by playing optimally a master can ensure victory.

The rules of the duel are as follows:

Evenius can only remove an **even** number ($2, 4, \dots$ etc) of stones on his turn, but not more than $\lfloor \frac{N}{2} \rfloor$ stones. Oddius can only remove an **odd** number ($1, 3, \dots$ etc) of stones on his turn, but not more than $\lfloor \frac{N}{2} \rfloor$ stones. Note that, here N represents the initial number of stones and $\lfloor x \rfloor$ denotes the floor function.

These two masters take turns removing stones from the pile, starting with a designated player. The master who is unable to make a valid move (can't remove any stone) loses the duel, and the other master is crowned the ruler of Numeria.

Both masters are strategists and will always play optimally to ensure their victory. Your task is to predict the outcome of this legendary battle and determine who will emerge victorious.

Input Description

The input consists of multiple lines. The first line contains an integer T ($1 \leq T \leq 10,000$) representing the number of test cases. Each of the next T lines contains two integers: N ($1 \leq N \leq 10^{16}$) denoting the initial number of stones in the pile and P representing the starting player, $P = 1$ represents Oddius and $P = 2$ represents Evenius.

Output Description

For each test case, print the test case number in the format "**Case X:** ", followed by the name of the master who will win the crown if both of the players play optimally. See the

sample Input/Output section for more clarity.

Samples

Input	Output
3	Case 1: Evenius
1 1	Case 2: Oddius
2 2	Case 3: Oddius
10 2	

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Ends in
02:50:13

THE CONTEST IS RUNNING

SUBMIT

Language

C++

-O2 -static -std=gnu++17

SELECT SOURCE

SUBMIT

MY SUBMISSIONS



No submissions yet!